GANDHINAGAR UNIVERSITY

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Gandhinagar Institute of Technology

Mathematics & Humanities Department

Discrete Mathematics (10050404)

Question Bank

Unit-1

SET THEORY

1. Let $U = \{1,2,3,4,5,6\}$, $A = \{1,2,3,4\}$, $B = \{4,5,6\}$, $C = \{2,4,6\}$. Find the following sets.
(a) A' (b) B' (c) $(A \cup B)'$ (d) $A - B$ (e) $(A \cup B) - C$ (f) $A \oplus B$.

2. Define power set. Determine power set $P(A)$ of the set $A = \{1,2,3,4\}$. How many elements are in $P(A)$?

3. Among 100 students, 32 study mathematics, 20 study physics, 45 study biology, 15 study mathematics and biology, 7 study mathematics and physics, 10 study physics and biology and 30 do not study any of the three subjects.

(a) Find the number of students studying all three subjects.
(b) Find the number of students studying exactly one of the three subjects.

4. Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

5. Prove that $(A \cup B)' = A' \cap B'$.

6. Prove that $A - (B \cap C) = (A - B) \cup (A - C)$.

7. Find how many integers between 1 and 60 that are not divisible by 2 not by 3 and not by 5. Also determine the number of integers divisible by 5, not by 2, not by 3.

8. Among 200 students in a class, 104 students got an 'A', in the first examination and 84 students got 'A' in second examination. If 68 students did not get an 'A' in either of the examination then How many students got 'A' in both the examination.

9. One hundred sportsmen were asked whether they ply which game: cricket, hockey, football. The results were :45 play cricket, 38 play hockey, 21 play football, 18 play cricket and hockey, 9 play cricket and football, 4 play football and hockey and 23 play none of these.
Draw a venn diagram that will show the result of the survey and determine the number of sportsmen who play: (a) Exactly one of the games (b) Exactly two of the games.

10. In a class of 55 students the number of studying different subjects are as below: maths 23, physics 24, chemistry 19, maths & physics 12, maths& chemistry 9, physics & chemistry 7, all three subjects 4. Then find the number of students who have taken: (a) at least one subject (b) exactly one subject
(c) Exactly three subject.

11. Prove that $A - (B - C) = (A - B) \cup (A \cap C)$.

12. If $A = \{1,2,3,4\}$, $B = \{2,3,4,5\}$, $C = \{3,4,5,6\}$ find A-B, A-(B-C).

13. Prove that $\emptyset = (A \cap B) \cap (A - B)$.

FUNCTION

1. If $A = \{x, y, z\}, B = \{a, b, c, d\}$. Check whether the following are function from A to B, if they are functions give the range of each, if not then give the reason.
   a) $f = \{(x, a), (y, b), (z, c)\}$
   b) $f = \{(x, a), (y, c), (z, b), (x, c)\}$
   c) $f = \{(x, d), (y, b)\}$
   d) $f = \{(x, a), (y, b), (z, d)\}$
2. If the function $f: R \to R$ defined by $f(x) = \begin{cases} 3x - 4, x > 0 \\ -3x + 2, x \leq 0 \end{cases}$.
   Determine $(a) f(0), f(2/3), f(-2)$ $\quad (b) f^{-1}(0), f^{-1}(-7), f^{-1}(2)$
3. Determine which of the following function $f: R \to R$ are one-one and which are onto R.
   a) $f(x) = x + 1$
   b) $f(x) = x^3$
   c) $f(x) = x + |x|$
4. Let : $N \to A$, be defined by $f(x) = 2x$, where A is the set of even integers. Check whether the function is one-one, onto or both. If f is one-one and onto then find f^{-1} .
5. Let f and g be two functions from R to R defined by $f(x) = x^2$ and $g(x) = x + 2$. Find fog and gof . Show that $fog \neq gof$. Also find $fog(1)$ and $gof(1)$.
6. Let $A = B = C = R$. Let $f: A \to B$ be defined by $f(x) = 3x + 1$ and $g: B \to C$ be defined by $(y) = {}^{y}/_{4}$. Verify that $(gof)^{-1} = f^{-1}og^{-1}$.
7. Define the following: 1 One One function. 2 Onto function 3. Bijective function. 4. Inverse of a function.
8. Let f, g, h be three functions from $R \to R$ as: $(x) = 2x^3 + 5$, $g(x) = \cos x$, $h(x) = x^3 - 1$. Find $ho(gof)$ and $(hog)of$. Are they equal?

COUNTING

1. In how many ways can the letters of the word *MONDAY* be arranged? How many of them begin with *M* and end with *Y?* How many of them do not begin with *M* but end with *Y?* **Ans. 720, 24, 96**
2. A man has 7 relatives, 4 of them are ladies and 3 gentlemen, his wife has 7 relatives and 3 of them are ladies and 4 gentlemen. In how many ways can they invite a dinner party of 3 ladies and 3 gentlemen so that there are 3 of man's relatives and 3 of wife's relative? **Ans. 485**
3. If ${}^{n}P_r = 72$, find the value of n. **Ans 9**
4. If ${}^{n}C_x = 56$ and ${}^{n}P_x = 336$, find n and x. **Ans.** $\boldsymbol{n = 8, x = 3}$
5. Out of 5 men and 2 women, a committee of 3 is to be formed. In how many ways can this be doe so as to include (i) exactly one woman (ii) at least one woman. **Ans. (i) 20 (ii) 25**
6. A house has 4 doors and 10 windows. In how many ways can a burglar rob the house by entering through a window and exiting through a door? **Ans. 40**
7. How many three-digit numbers can be formed without using the digits 1, 1, 3? **Ans. 216**
8. How many numbers are there between 100 and 1000 such that 7 is in the unit's places? **Ans. 90**
9. How many permutations can be made out of the letters of the word *TRIANGLE*? How many of these will begin with T and end with E? Also find how many permutations will begin with T but will not end in E? **Ans. 40320, 720, 4320**
10. In how many ways can the letters of the word *COMBINE* be arranged so that (i) the vowels are never separated (ii) all the vowels never come together (iii) vowels occupy only the odd places? **Ans. 720,4320,576**

11. How many arrangements can be made with the letters of the word *MATHEMATICS*? In how many of them the vowels occur together? **Ans. 4989600, 120960**

12. There are 6 gentlemen and 4 ladies to dine at a round table. In how many ways they can seat themselves so that no two ladies are together? **Ans. 43200**

13. In how many ways can 10 people sit around a table so that all shall not have the same neighbours in any two arrangements? **Ans.** $\frac{1}{2}(9!)$

14. A committee of 11 members sits at a round table. In how many ways can they be seated in if the president and the secretary choose to sit together? **Ans. 2 (9!)**

15. (a) What is the coefficient of x^6y^3 in $(x+y)^9$?

(b) What is the coefficient of x^5y^7in $(x-2y)^{12}$?

(c) What is the coefficient of x^3y^2 in $(3x-y)^5$?

(d) What is the coefficient of $x^2y^3z^2$ in $(x+y+z)^7$?

(e) What is the coefficient of $x^3y^2z^5$in $(x+y+z)^{10}$?

(f) What is the coefficient of x^2yz in $(2x-y+z+1)^7$?

Ans. (a) 84 (b) -101376 (c) 2700 (d) 210 (e) 2520 (f) -280

16. Show that 7 colours are used to paint 50 bicycles, at-least 8 bicycles will be of same colour.

17. In how many ways letter 'ORGANISE' can be arranged in such a way that all vowels always come together.

18. In how many ways can you take 5 cards, with at-least two aces, out of well shuffled pack of 52 cards.

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Unit-2

PREDICATE LOGIC

1. Write negations for each of the following:
   a) Real number $x, if\ x > 3\ then\ x^2 > 9$.
   b) $x \in R\ if\ x(x+1) > 0$ then $x > 0\ or\ x < -1$.
   c) Integers $a, b\ and\ c\ if\ a - b$ is even and $b - c$ is even, then $a - c$ is even.

2. Rewrite the following argument using quantifiers, variables and predicate symbols:
   a) All birds can fly.
   b) Not all birds can fly
   c) Some med are genius.
   d) Some numbers are not rational.
   e) There is a student who likes mathematics but not Geometry.

3. Remove all negations of the quantifiers of following expression:

$$\sim\forall y(\exists\ x\ P\ (x,y)\ \wedge \sim\ \forall x\ Q(x,y))$$

4. Translate each of the statement into symbols, using quantifiers, variables and predicate symbols:

   Let $P(x)$: x can speak Tamil and $Q(x)$: x knows the language $C++$

   a) There is a student who can speak Tamil and who knows $C++$
   b) There is a student who can speak Tamil but does not know $C++$
   c) Every student either can speak Tamil or knows $C++$
   d) No student can speak Tamil or knows $C++$

5. Use predicate calculus to prove the following agreement:
   a) All mammals are animals. Some mammals are two-lagged. Therefore, some animals are two-legged.
   b) All animals are mortal. All human being are animals. Therefore, all humans are mortal.
   c) No human being are quadrupeds. All women are human beings. Therefore, no women are quadrupeds.

PROPOSITIONAL LOGIC

1. For the statements p and q, Define Conjunction and Disjunction. Also write their truth tables.

2. Define concept of contrapositive of implication $p \rightarrow q$ and write its truth table.

3. Express the contrapositive, converse, inverse and negative forms of the following conditional statement. 'if x is rational, then x is real'.

4. State inverse and converse of the following implications.
   (a) If I am fine, then I go to gym.
   (b) If I am the director of an institute, then I will go by car.
   (c) If you work hard, then you can earn money.

5. If p, q and r are three statements, using truth table show that
   (a) $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$

(b) $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$

6. Using truth table prove the absorption laws

(a) $p \wedge (p \vee q) = p$

(b) $p \vee (p \wedge q) = p$

7. Define tautology. Show that $\sim (p \wedge (\sim q))$ is a tautology.

8. Show that $(p \rightarrow q) \leftrightarrow (\sim p) \vee q$ is a tautology.

9. If the statements, p: I read book and q: I go for swimming then find

(i) $p \vee q$ (ii) $p \wedge q$ (iii) $(\sim p) \vee q$ (iv) $p \wedge (\sim q)$

10. Construct truth table for $(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$.

11. Construct truth table for $(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$.

12. Construct truth table for $(p \wedge q) \wedge (\sim (p \vee q))$.

13. For the statements p and , Show that $p \leftrightarrow q$ is logically equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$.

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Unit-3

RELATION

1. Let $A = \{2,3,4\}$ and $B = \{3,4,5\}$. List the elements of each relation R defined below and the domain and range.

a) $a \in A$ is related to $b \in B$ that is $a\ R\ b$ if and only if $a < b$.

b) $a \in A$ is related to $b \in B$ that is $a\ R\ b$ if a and b are both odd numbers.

2. Give an example of a relation which is:

a) Reflexive and transitive but not symmetric

b) Symmetric and transitive but not reflexive

c) Reflexive and symmetric but not transitive

3. If R be a relation in the set of integers Z defined by $R = \{(x, y): x \in Z, y \in Z, (x - y)$ is divisible by $6\}$ Then prove that R is an equivalence relation.

4. Given $A = \{1,2,3,4\}$, $B = \{a, b, c\}$ and $C = \{x, y, z\}$. Let R and S are the following relations from A to B and B to C, respectively $R = \{(1, b), (2, a), (2, c)\}$ and $S = \{(a, y), (b, x), (c, y), (c, z)\}$

a) Determine the matrix and graph of the relation R , S, R^{-1}, S^{-1}, and $R \circ S$.

b) Determine inverse R^{-1} and S^{-1}.

c) Find the range and domain of R^{-1} and S^{-1}.

5. Consider the following relation on $\{1,2,3,4,5,6\}$. $R = \{(i, j): |i - j| = 2\}$
Is $'R'$ transitive? Is $'R'$ reflexive? Is $'R'$ symmetric?

6. Let R be a binary relation defined as $R = \{(a, b) \in R^2 : (a - b) \leq 3\}$
Determine whether R is reflexive, symmetric, antisymmetric and transitive.

7. Let $A = \{1,2,3,4,5\}$ and $R = \{(1,2), (1,1), (2,1), (2,2), (3,3), (4,4), (4,5), (5,4), (5,5)\}$ be equivalence relation on A. Determine the partitions corresponding to R^{-1} , if it is an equivalence relation.

8. Let $S = \{1,2,3,4,5,6,7,8,9\}$ and let $\sim$ be the relation on A$\times$ A defined by $(a, b) \sim (c, d)$ whenever $a + d = b + c$

a) Prove that $\sim$ is an equivalence relation.

b) Find equivalence class of $(2,5)$ or $[(2,5)]$.

9. Consider the set $\boldsymbol{Q}$ of rational numbers with the usual order $\leq$ and D is the subset of $\boldsymbol{Q}$ defined by D $=\{y \mid y \in \mathbf{Q}$ and $4<y^2<13\}$

(a) Is D bounded above or below?

(b) Does $\sup(D)$ or $\inf(D)$exist? If yes, then find them.

10. If R be a relation in the set of integers Z defined by
$R = \{(x, y): x \in Z, y \in Z, (x - y)$ is multiple of $3\}$ Show that it is an equivalence relation. What is the equivalence class of 0? How many equivalence classes are there?

11. Draw the directed graph that represents the relation
$R = \{(1,1), (2,2), (1,2), (2,3), (3,2), (3,1), (3,3)\}$ on $X = \{1,2,3\}$

12. $A = \{1,2,4,8\}$ $a\ R\ b$ if and only if $a|b$ (a divides b) then find the digraph of relation.

13. $A = \{1,2,3,4\}$ the relation matrix is $M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$. Draw its graph.

14. Consider a relation R defined on $A = \{1,2,3\}$ whose matrix representation is given below. Determine its inverse R^{-1} and complement R'.

$$M_R = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

PARTIAL ORDERING

1. Show that the relation $\leq$ is a POSET on the set of integers.

2. Shoe that the relation $>$ is not a POSET on the set of integers.

3. Shoe that the set of positive integers under divisibility forms a POSET.

4. Let $x = \{1,2,3,4,5,6\}$, then / is a POSET on x. draw the Hasse diagram of (X,/).

5. LET $A = \{1,2,3,4\}$ and consider the relation $R = \{(1,1), (2,1), (2,2), (3,1), (3,3), (3,4), (4,4)\}$ show that R is a POSET and also draw its Hasse diagram.

6. Consider the divides relation on each of the following sets S. draw the Hasse diagram for each relation then find

a) All minimal and maximal element where $S = \{2,3,5,30,60,120,180,360\}\}$

b) Greatest and least element where $S = \{1,2,3,4,6\}$.

7. Find the greatest lower bound and the least upper bounds of the sets $\{3,9,12\}$ and $\{1,2,4,5,10\}$ if they exist, in POSET$(Z^+, 11)$.

8. Answer the following questions concerning the poset $\{\{1\}, \{2\}, \{4\}, \{1,2\}, \{1,4\}, \{2,4\}, \{3,4\}, \{1,3,4\}, \{2,3,4\}, \subseteq\}$

a) Find the maximal and minimal elements.

b) Find the upper bounds of $\{\{2\}, \{4\}\}$ and the least upper bound, if it exists.

c) Find the lower bounds of $\{1,3,4\}$ and the greatest lower bound of $\{1,3,4\}$.

9. Prove that pentagonal lattice is not modular.

10. Show that in a complemented, distributive lattice, the followings are equivalent.

a) $a \preccurlyeq b$

b) $a \wedge b' = 0$

c) $a' \vee b = 1$

d) $b' \preccurlyeq a'$.

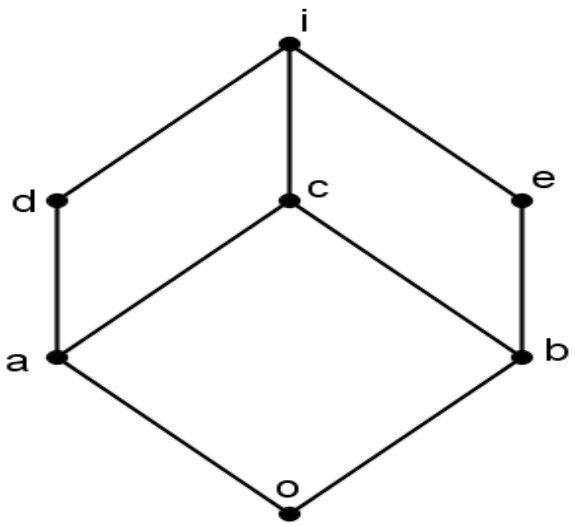
11. Observing the following diagrams

(i) which of the following are the sublattices of L: $L_1 = \{0, a, b, i\}, L_1 = \{0, a, e, i\}, L_1 = \{a, c, d, i\} L_1 = \{0, c, d, i\}$

(ii) Is L is distributive?

(iii) Find the complements of a, b and c if they exist.

(iv) Is L a complemented lattices?

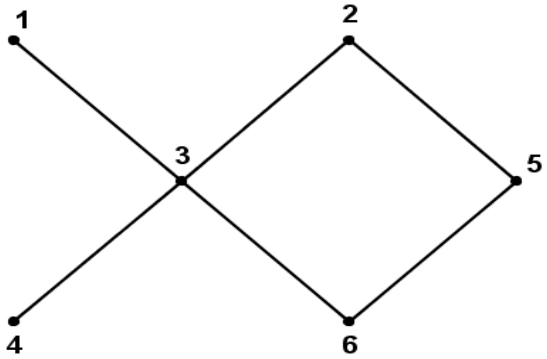


12. Let $M = \{2,3,4, \dots \dots\}$ and let $M^2 = M \times M$ be ordered as follows:

$$(a,b) \prec (c,d) \; if \; a|c \;\; and \;\; b < d$$

Find all minimal and maximal elements of $M \times M$.

13. Let M = {1,2,3,4,5,6} be ordered as in below Fig. Find all minimal and maximal elements of M. Does M have a first or last elements?



14. Find the solution of recurrence relation

(i) $a_n = 3a_{n-1} - 10a_{n-2} \qquad a_0 = 5, a_1 = 11$

(ii) $a_n = 11a_{n-1} - 39a_{n-2} + 45a_{n-3}$

15. Consider the set A = {1, 2, 3, 9, 18}. Let R be the relation a ≤ b iff a|b on A. Draw the directed graph and the Hasse diagram of R.

16. Consider the set $\boldsymbol{Q}$ of rational numbers with the usual order $\leq$ and D is the subset of $\boldsymbol{Q}$ defined by

$D = \{(-1)^x - \frac{1}{x}$ such that x ε N}

(i) Is D bounded above or below?

(ii) Does sup(D) or inf(D)exist? If yes, then find them.

17. Consider the set A = {4, 5, 6, 7}. Let R be the relation ≤ on A. Draw the directed graph and the Hasse diagram of R.

Unit-4

GROUP THEORY

1. Let Q_+ be the set of all positive rational numbers and $'*'$ a binary operation on Q_+ defined by $a * b = \frac{ab}{3}$ is an abelian group.
2. Let $R = \{0°, 60°, 120°, 180°, 240°, 300°\}$ and $* =$ binary operation, so that for a and b in R $a * b$ is overall angular rotation corresponding to successive rotations by a and then by b. Show that $(R,*)$ is a group.
3. if * is a binary operation in R defined by $a * a = a + b - ab \quad \forall\, a, b \in R$ then Show that the inverse of $a \neq 1$ is $\frac{a}{a-1}$.
4. G is a group and there exists two relatively prime positive integers m and n such that $a^m b^m = b^m a^m$ and $a^n b^n = b^n a^n$ for all $a, b \in G$. Prove that G is an abelian group.
5. Prove that $G = \{1, \omega, \omega^2\}$ is a group with respect to multiplication where $1, \omega, \omega^2$ are cube roots of unity.
6. Let G be the set of all non-zero real numbers and let $a * b = \frac{ab}{2}$ Show that $(G,*)$ is an abelian group.
7. Let G be a group, for a fixed element of G, Let $G_x = \{a \in G; ax = xa\}$. Show that G_x is a subgroup of G for all $x \in G$.
8. Show that the set of four transformations f_1, f_2, f_3 *and* f_4 on the set of complex numbers be defined by $f_1(z) = z, f_2(z) = -z, f_3(z) = \frac{1}{z}, f_4(z) = \frac{-1}{z}$ Forms a finite abelian group with respect to the binary operation as the composition of product of two functions.
9. Four fourth roots of unity form a cyclic group $G = \{1, -1, i, -i\}$.
10. Let G be a group with identity e. Show that function $f: G \to G$ defined by $f(a) = e$ for all $a \in G$ is a homomorphism.
11. Show that the set $G = \{0,1,2,3,4\}$ is an abelian group with respect to addition modulo 5.
12. Prove that the set $\{1,2,3,4,5,6\}$ of order 6 is a finite abelian group of order 6 under multiplication modulo 7.
13. Prove that the set $G = \{a + \sqrt{2}b : a, b \in Q\}$ is a group with respect to addition.

14. If $(G,*)$ is a group then $(a*b)^{-1} = b^{-1}*a^{-1}$ for all $a,b \in G$

15. Let $H = \{1,-1\}$ and $G = \{1,-1,i,-i\}$ then $(H,*)$ is a sub group $(G,*)$

16. The groups $(Z,+)$ and $(nZ,+)$, $n \in Z$ are isomorphic to each other.

17. State and prove that Cayley's Theorem.

18. State and Prove the Fundamental Theorem on Homomorphism.

19. If $A = \{1,2,3,4,5,6\}$ compute $(5\ 6\ 3)°(4\ 1\ 3\ 5)$

20. Show that $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 4 & 3 & 6 & 5 & 8 & 7 & 2 \end{pmatrix}$ is odd.

21. Set M of 2×2 matrices where entries of matrices are real numbers form a ring with unity with respect addition (+) and multiplication (.) of matrices.

22. Consider the set $(C,+,.)$ where C is the set of complex numbers and "+" and "." are ordinary addition and multiplication operation. Show that $(C,+,.)$ is a ring.

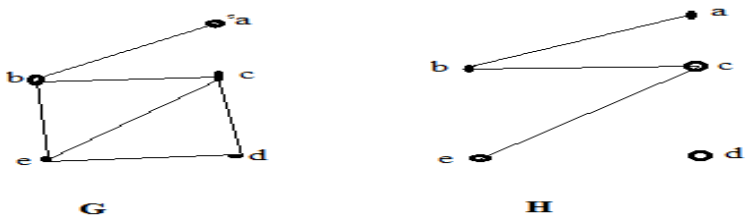
23. Show that $(F,+,.)$ is a field where F is set of all rational numbers and "+" and "." are ordinary addition and multiplication operators.

24. Show that $R = \{a+\sqrt{2}b : a,b \in I\}$ for the operation $+,\times$ is an integral domain but not a field.

Unit-5

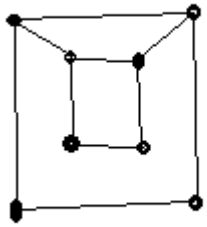
GRAPHS, REACHABILITY, CONNECTEDNESS

1. How many nodes are necessary to construct a graph with exactly 8 edges in which each node is of degree 2
2. Draw a simple graph with 6 nodes all of degree 2 or greater and with at least 2 nodes of degree 3.
3. Draw a graph with 4 nodes and 7 edges.
4. Is there exist a regular graph of degree 5 on 9 vertices? Give reason
5. Determine the number of edges in a graph with 6 nodes, 2 of degree 4 and 4 of degree 2. Draw two such graphs.
6. Draw a complete bipartite graph which is not regular.
7. Draw a graph which is regular but not bipartite.
8. How many edges have each of the following graphs $(1)K_7$ $(2)K_{3,7}$
9. If G is a graph and H is a subgraph then draw complement of H



10. Is $K_6$is 1-factorable graph? Then draw 1-factor non isomorphic graphs.

11. Find following graphs are isomorphic or not?



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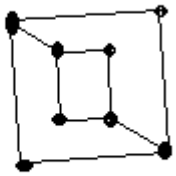

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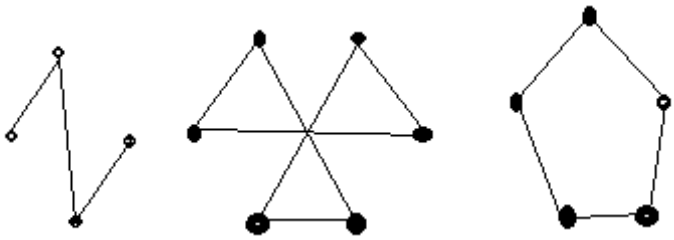
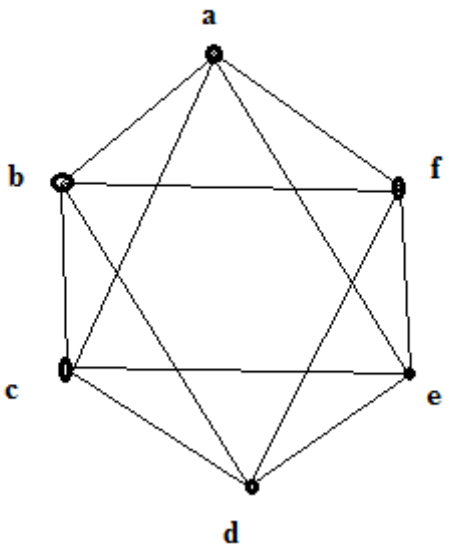
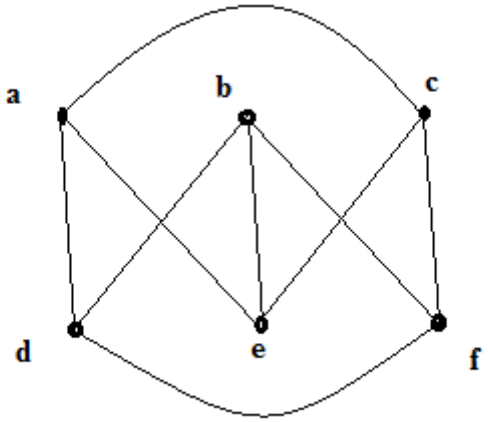
12. Draw complements of given graphs which are subgraphs of complete graphs

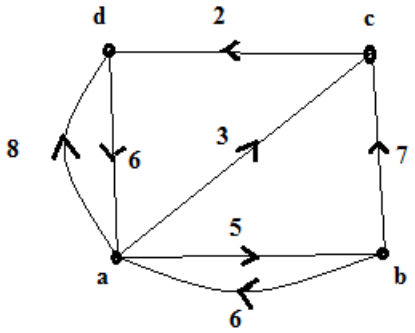


13. Draw all non-isomorphic simple graphs with 5 vertices.
14. Whether K_4 & $K_{2,2}$ are isomorphic?
15. Draw a graph which contains an Eulerian circuit
16. Draw a graph which contains an Eulerian path but does not contain an Eulerian circuit.
17. Find under what conditions $K_{m,n}$ the complete bipartite graph will have an Eulerian circuit
18. Draw a planar representation of each graphs for the following figure if possible





19. Find the shortest path between each pair of vertices for a simple digraph using Warshall's algorithm.



20. If $G = (V, E)$ is a simple connected planar graph then $e \leq 3v - 6$ where e is total number of edges and v is total number of vertices in the graph G.

21. Draw all non-isomorphic trees with six and seven points.

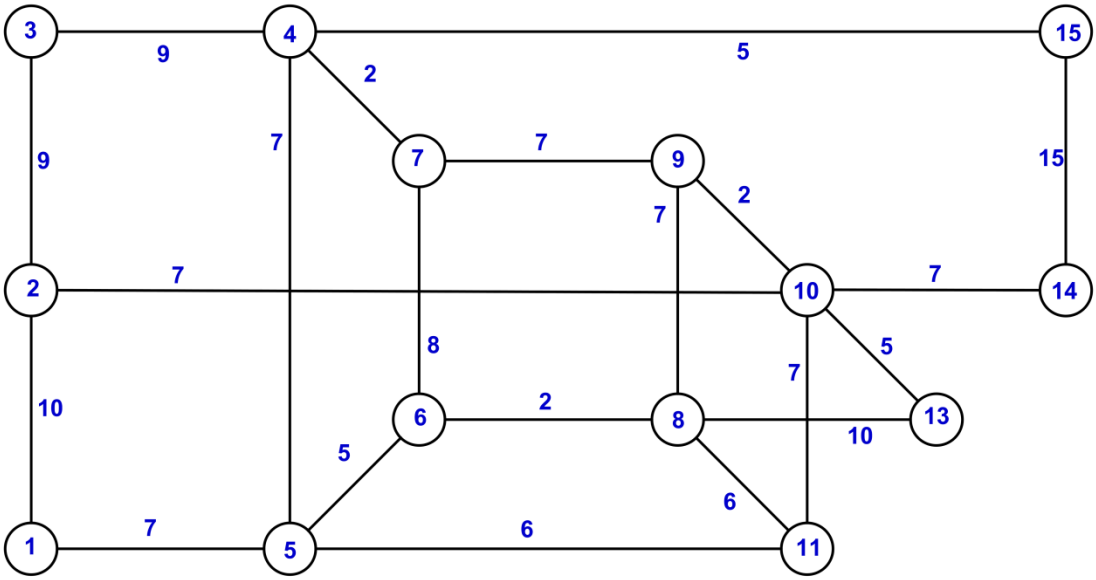
22. Show that it is possible to draw a tree with 10 vertices which has vertices either of degree 1 or of degree 3. Draw the tree. Is it possible to draw the same type of tree with 11 vertices?

23. A telephone network is established among 100 people. Information received by the first person is passed along to the 99 others as follows; the first person calls exactly 3 people, and each of these people call 3 others and so on till there are no others to call. If each call taken 5 minutes, then how long does it takes for message to be relayed from first person to receive the message to everyone else? How many people make no calls?

24. What is the total number of nodes in a full binary tree with 20 leaves?

25. 19 lamps are to be connected to a single electrical outlet, using extension chords, each of which has 4 outlets, find the number of extension chords needed and draw the corresponding tree.

26. For the following set of weights construct an optimal binary prefix code. For each weight in the set give corresponding code word. 5,7,8,15,35,40.



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